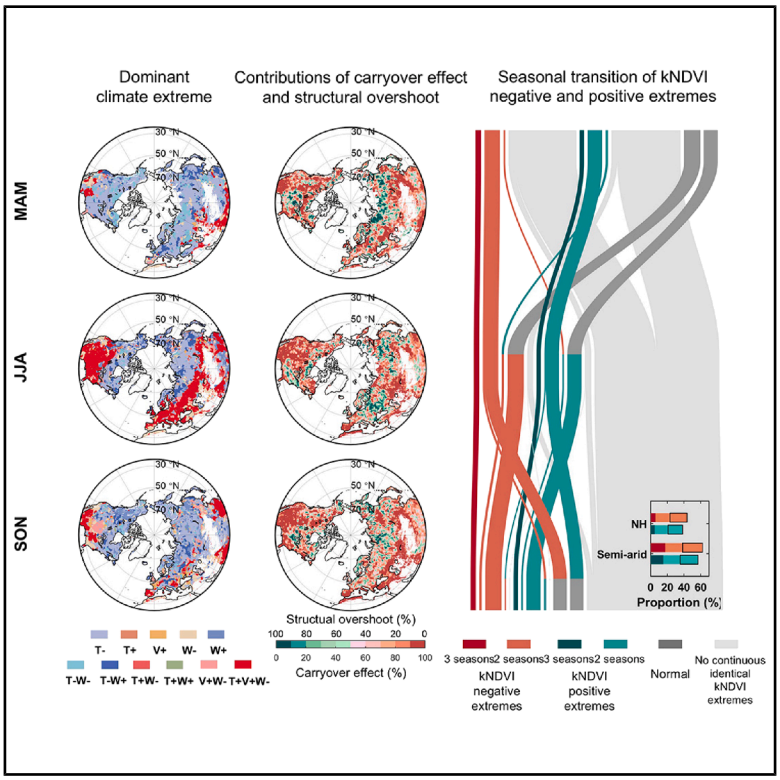
One Earth

Article

Concurrent climate extremes and biological carryover effects dominate severe seasonal reductions in northern vegetation growth

Graphical abstract



Authors

Yiting Luo, Hui Yang, Hao Xu, Chris Huntingford, Rene Orth, Xiangyi Li, Josep Peñuelas

Correspondence

yang_hui@pku.edu.cn (H.Y.), xuhao@iiasa.ac.at (H.X.)

In brief

Luo et al. quantified the drivers of extreme negative vegetation growth anomalies across different seasons, showing that about 60% were attributable to concurrent climate extremes, while over 10% resulted from vegetation preconditions in the previous season through biological carryover effects. These carryover effects extend the duration of severe seasonal vegetation reductions, particularly in semi-arid regions.

Highlights

- Seasonal vegetation extremes are driven by climate extremes and past plant states
- Dominant climatic drivers of vegetation extremes vary by both biome and season
- Biological legacy extends the duration of severe seasonal vegetation reductions
- Spring and summer vegetation extremes often persist due to weak recovery capacity

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